158.888 Information Technology Professional Project

Reflection and Peer Review Report on

Real-Time Parking Occupancy Monitoring and Area-Level Prediction Using Spatio-Temporal Analytics

Presented by

Group 2

S M Amit Hasan – 25004146

Project Coordinated By

Associate Professor Anu Mathrani

a.s.mathrani@massey.ac.nz

&

Ms Binglan Han

b.han1@massey.ac.nz

Date of Submission: 5th June 2026

Massey University

College of Sciences

Albany Campus, Auckland, New Zealand

1. Reflection on the Project

This project significantly enhanced my understanding of machine learning methodologies, feature engineering techniques, API integration processes, dashboard development, and smart city analytics. I gained valuable practical experience in integrating Python-based analytical workflows with Power BI visualisation platforms to develop intelligent transportation system applications. The project also strengthened my problem-solving, data integration, and predictive modelling skills within real-world transportation analytics contexts.

One of the most valuable aspects of the project was the opportunity to work with real-world transportation datasets and live Auckland Transport parking feeds. Unlike theoretical classroom exercises, this project required practical implementation using real operational data collected from multiple cities including Auckland, Melbourne, Brisbane, and Geelong. Working with heterogeneous datasets exposed me to several realistic challenges such as inconsistent timestamp formats, missing values, occupancy normalization, varying parking structures, and different occupancy measurement techniques. Addressing these issues improved my understanding of data pre-processing and analytical workflow management.

The project also provided practical exposure to machine learning forecasting workflows and predictive analytics implementation. I was actively involved in machine learning forecasting implementation, feature engineering, prediction validation, and forecasting optimization. Through this process, I learned how occupancy-history variables, lag-based features, rolling occupancy statistics, temporal indicators, and weather-related attributes contribute to forecasting accuracy within intelligent transportation systems.

One of the major lessons learned during the project was the importance of proper validation strategies in time-series forecasting. Initially, random train-test splitting methods were used during experimentation. However, after further analysis and testing, the team recognized that chronological validation techniques produced more realistic forecasting performance because they better simulated operational prediction scenarios. Understanding the importance of forecasting validation methods significantly improved my knowledge of predictive analytics and machine learning implementation.

Another important learning outcome was understanding how machine learning forecasting can support smart city management and urban transportation planning. The final implementation demonstrated how predictive analytics, real-time API integration, dashboard visualization, and GIS-based hotspot mapping can work together within a unified smart parking intelligence framework. Observing how real-time operational dashboards can support parking demand analysis and transportation decision-making gave me a deeper appreciation for intelligent transportation systems and smart city technologies.

The project also strengthened my understanding of collaborative software development and interdisciplinary project management. Since the implementation involved machine learning forecasting, Power BI dashboard development, GIS visualization, and operational analytics, effective coordination between team members became extremely important. Regular communication, collaborative testing, and iterative refinement helped ensure successful integration between forecasting outputs, dashboard visuals, and spatial intelligence components.

One of the most challenging aspects of the project involved integrating forecasting outputs with dashboard visualization systems. During implementation, we encountered several practical issues involving occupancy categorization, inconsistent location naming, dashboard filtering behaviour, visualization synchronization, and real-time prediction updates. Participating in debugging and refinement activities improved my problem-solving abilities and practical technical experience.

If I were to repeat the project again, I would spend more time designing a structured workflow and standardized data architecture during the early implementation stages. More structured planning could help reduce later integration challenges and improve development efficiency. I would also recommend implementing automated pre-processing and dashboard refresh pipelines earlier in the project lifecycle.

Overall, this project provided highly valuable academic and practical experience in predictive analytics, machine learning workflows, GIS-based visualization, dashboard analytics, and intelligent transportation systems. The project strengthened both my technical and collaborative skills while improving my understanding of how AI-driven technologies can support smart city operations and urban transportation management.

2. Peer Review

My primary role in the project involved machine learning implementation, forecasting model development, feature engineering, data pre-processing, and prediction pipeline integration. I mainly handled occupancy forecasting models, validation strategies, API-based forecasting integration, and machine learning experimentation. I also contributed to feature engineering workflows, prediction evaluation, occupancy categorization logic, and forecasting refinement activities.

Throughout the project, I actively participated in debugging, testing, and integration activities associated with forecasting outputs and dashboard synchronization. I contributed to ensuring that the machine learning prediction results could be successfully integrated into the Power BI dashboards and operational visualization framework. I was also involved in refining occupancy forecasting workflows and improving prediction consistency using iterative testing methods.

Shovan Barai mainly contributed to Power BI dashboard development, visualization design, KPI implementation, occupancy categorization, and dashboard customization. His contribution significantly improved the visual presentation and operational reporting quality of the project. He worked extensively on dashboard styling, forecasting visualization layouts, interactive filtering, and KPI card implementation. His work helped transform the analytical outputs into user-friendly operational dashboards suitable for smart parking analytics.

Amit Chakrabarty mainly focused on GIS-based hotspot analysis, documentation support, dashboard testing, and spatial visualization activities. His contribution helped improve the spatial intelligence and geographic visualization aspects of the project. He was also involved in testing dashboard functionality, reviewing occupancy categorization consistency, and supporting report preparation activities.

The group allocated tasks according to technical strengths and individual expertise. Regular communication and collaborative coordination were maintained throughout implementation, testing, visualization refinement, and report preparation phases. Team discussions helped ensure that forecasting outputs, dashboard visualizations, and GIS-based hotspot analytics worked consistently together within the final smart parking intelligence platform.

One of the strengths of the group collaboration was the ability to integrate different technical domains into a unified operational framework. Since the project combined machine learning forecasting, Power BI analytics, GIS mapping, API integration, and transportation analytics, close coordination between team members became extremely important. Frequent meetings and collaborative testing sessions helped resolve implementation challenges and ensured that all project components aligned properly.

Overall, the collaborative structure of the project was effective and successful. Each group member contributed according to their specialization area while also supporting integration and testing activities across the project lifecycle. The teamwork and collaborative problem-solving process significantly improved the quality of the final implementation and helped ensure successful completion of the project objectives.